GGG-H-35a
FEBRUARY 28, 1957
SUPERSEDING
Int. Fed. Spec. GGG–H–0035 (Navy–Ships)
February 2, 1954

FEDERAL SPECIFICATION

HAMMER, HAND; WELDERS', CHIPPING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers hand hammers used by welders for chipping welds and welding slag.

1.1.1 Federal specification coverage.—Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.*—Hammers shall be of the following types and classes, as specified (see 6.2):

Type I.—Straight peen and punch point.
  Class 1.—Single bevel.
  Class 2.—Double bevel.

Type II.—Cross peen and punch point.
  Class 1.—Single bevel.
  Class 2.—Double bevel.

Type III.—Straight peen and cross peen.

Type IV.—Straight peen and brush.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specification and standards of the issues in effect on date of invitation for bids, form a part of this specification:

Military Specification:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Federal Standard:

Federal Test Method Standard No. 151—Metals; Test Methods.

Military Standard:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Nongovernmental:

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies of the Consolidated Freight Classification Rules and Container Specifications should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations herein are descriptive and not restrictive, and are included for the convenience of requisitioning and purchasing officers and manufacturers. They are not intended to preclude the purchase of tools which are otherwise in accordance with this specification.

3.2 Material. — The hammers shall have steel chipping heads with steel, wood or plastic handles (see 3.3.2).

3.2.1 *Steel.*—The steel shall be as specified herein.

3.2.2 *Wood.*—The wood shall be high quality, straight-grained, sound, thoroughly seasoned wood of any species which is generally commercially furnished for the purpose intended. The wood shall be free from defects which would impair its durability.

3.2.3 *Plastic.*—The plastic shall have a lustrous finish, shall be tough and fire resistant and shall possess good impact properties. Plastic handles shall be capable of passing the tests specified in 4.4.3.

3.3 Construction.—Hammers shall be fabricated so as to be strong, durable, and permit chipping of all types of welds and welding slag. Hammers shall consist of a head and a handle as specified herein. The handle shall be positioned at right angles on the chipping head so as to provide a balanced tool.

3.3.1 *Chipping head.*—The chipping head shall be of hot-forged steel of hexagonal, octagonal, or square cross section.

3.3.1.1 *Work ends.*—The work ends of the chipping head shall be suitable for the use intended. The cutting edge of the chisel ends and the points of the punch ends shall not crack, fracture, chip, flatten, or show undue wear when subjected to the test specified in 4.4.1. Chisel-shaped ends shall be single or double beveled as specified herein.

3.3.1.1.1 *Hardness.*—The work ends of the chipping head shall be heat-treated to indicate a Rockwell hardness of not less than 52 nor more than 60 on the C scale.

3.3.2 *Handles.* — The handles shall be welded to, or be an integral forged part of the head. Unless otherwise specified in the contract or order, handles shall be of one of the following styles at the option of the contractor (see 6.1 and figures 1 to 6, inclusive):

Style A—Flat steel handle.
Style B—Plastic handle.
Style C—Round wood handle.
Style D—Coiled steel handle.
Style E—Forged steel handle.

3.3.2.1 *Style A, flat steel handle.*—Style A handles shall be of round, mild-steel bar so formed as to present a flat steel handgrip. A ring of the same diameter bar shall be securely welded to the handle at right angles at a point just above the grip of the handle. Style A handles shall be similar in design, and shall conform to the dimensions of figure 1 (see 3.6.1).

3.3.2.2 *Style B, plastic handle.*—Style B handles shall consist of a round mild-steel bar with a securely fastened fluted plastic handgrip. Plastic handles shall comply with the test requirements specified in 4.4.3, shall

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conform to table I, and shall be similar in design to figure 2 (see 3.6.2).

TABLE I.—*Styles B, C, and D; plastic handle, round wood handle, and coiled steel handle*

Largest diameter of handle grip D ± 1/16	Length of handle grip E ± ¼	Diameter of mild steel bar F ± 1/16
Inches	*Inches*	*Inch*
1¼	5	⅝

3.3.2.3 *Style C, round wood handle.*—Style C handles shall consist of a round mild-steel bar with a securely fastened round, metal-ferruled, wooden handgrip, shall conform to table I, and shall be similar in design to figure 4 (see 3.7.2).

3.3.2.4 *Style D, coiled steel handle.*—Style D handles shall consist of a round mild-steel bar with a securely fastened spring wire spiral handgrip, shall conform to table I, and shall be similar in design to figure 5 (see 3.8).

3.3.2.5 *Style E, forged steel handle.*—Style E handles shall be of forged steel incorporating a safe, comfortable handgrip. Style E handles shall conform to the dimensions of figure 6 (see 3.9). Hammers furnished with style E handles shall not exceed one pound in weight.

3.3.3 *Welds.*—Welds consistent with good commercial techniques shall be of mild steel and shall be free from defects which would impair the weld efficiency. Silver brazing will not be acceptable.

3.4 Marking for identification.—Each hammer shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.5 Finish.—The head and handle of the hammers shall be free from burrs, roughness, and excessive oxide scale, and shall have a lacquered, enameled, painted, or plated finish. The work ends, including taper, shall be smoothly ground and shall have a natural finish.

3.6 Type I, straight peen and punch point.—Type I hammers shall consist of a head with two chipping ends, and a handle. One chipping end shall be chisel-shaped with the cutting edge directly in line with the longitudinal axis of the head. The other chipping end shall be of square cross section tapered to a punch point having an included angle of 60 degrees plus or minus 5 degrees. The head shall have the cutting edge of the chisel end parallel to the handle. Type I hammers shall conform to table II.

TABLE II.—*Type I, type II, and type III; straight peen and punch point, cross peen and punch point, and straight peen and cross peen*

Over-all length A ± ¼	Head	
	Length B ± ¼	Width (distance across flats and chipping edge) C ± 1/16
Inches	*Inches*	*Inch*
10	6	⅝

3.6.1 *Class 1, single bevel.*—Class 1 hammers shall have the chisel-shaped chipping end of the single bevel shear type at an angle from the horizontal of 45 degrees plus or minus 5 degrees, and shall be similar to figure 1.

3.6.2 *Class 2, double bevel.*—Class 2 hammers shall have the chisel-shaped chipping end of the double bevel shear type with an included angle of 60 degrees plus or minus 5 degrees, and shall be similar to figure 2.

3.7 Type II, cross peen and punch point.—Type II hammers shall conform to 3.6 except that they shall have the cutting edge of the chisel end at a right angle to the handle.

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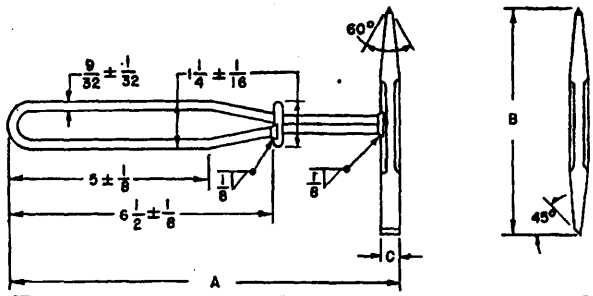


FIGURE 1.—Type I, class 1, straight peen and punch point, single bevel (with style A, flat steel handle).

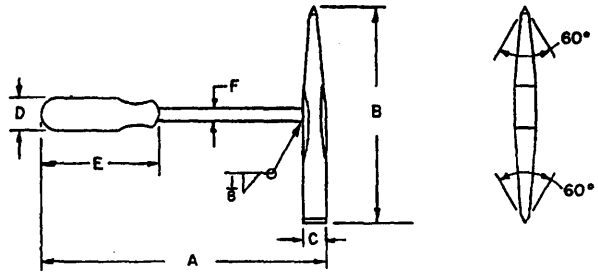


FIGURE 2.—Type I, class 2, straight peen and punch point double bevel (with style B, plastic handle).

3.7.1 *Class 1, single bevel.*—Class 1 hammers shall conform to 3.6.1 except that they shall be similar to figure 3.

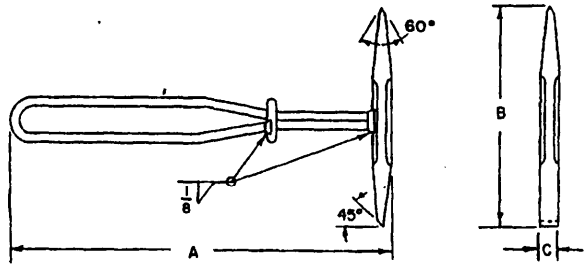


FIGURE 3.—Type II, class 1, cross peen and punch point, single bevel (with style A, flat steel handle).

3.7.2 *Class 2, double bevel.*—Class 2 hammers shall conform to 3.6.2 except that they shall be similar to figure 4.

3.8 Type III, straight peen and cross peen. —Type III hammers shall consist of a head, with two double beveled (60 degrees included angle) chisel-shaped chipping ends, and a handle. One cutting edge shall be parallel to the handle and the other cutting edge shall

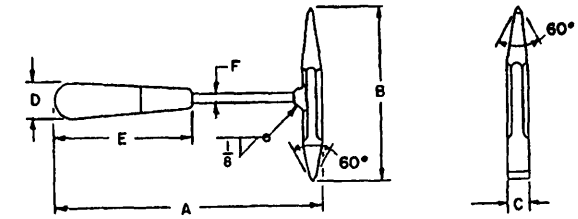


FIGURE 4.—Type II, class 2, cross peen and punch point, double bevel (with style C, round wood handle).

be at a right angle to the handle. Both cutting edges shall be directly in line with the longitudinal axis of the head. Type III hammers shall conform to table II and shall be similar to figure 5.

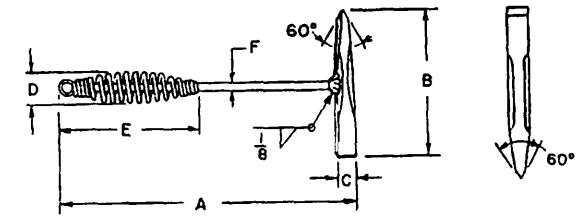


FIGURE 5.—Type III, straight peen and cross peen (with style D, coiled steel handle).

3.9 Type IV, straight peen and brush. — Type IV hammer heads shall consist of a double beveled (60-degree, included angle) chisel-shaped chipping end, with the cutting edge of the chipping end parallel to the handle, and a wire brush. The wire brush backing shall be made of wood (see 3.2.2), and the block shall be slightly curved so that the faggots protrude from a longitudinal convex surface. The brush shall be securely fastened to the head end of the handle and in such a manner that it may be readily and easily replaceable. The brush stock shall be of round medium hard steel wire. The faggots of wire shall be securely fastened into the backing and be of uniform length. Type IV hammers shall conform to table III and shall be similar to figure 6.

3.10 Workmanship. — Workmanship shall be of the highest grade and equal in every respect to good commercial practice. Hammers shall be free from laps, burrs, cracks,

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TABLE III.—*Type IV, straight peen and brush*

Brush										Over-all length A ± ¼	Chipping head	
Block			Number of rows		Wire		Holes per row ± 2	Wires per hole ± 2	Diameter of holes ± 1/32		Length B ± ¼	Width (distance across flats and chipping edge) C ± 1/16
Width ± ⅛	Length ± ¼	Thickness ± ¼	Minimum	Maximum	Diameter ± 0.002	Length of faggots (outside) ± ¼						
Inch	*Inches*	*Inch*			*Inch*	*Inches*			*Inch*	*Inches*	*Inches*	*Inch*
⅞	4½	1	3	4	0.012	1¼	16	32	3/32	10½	4	⅝

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and other defects which may impair their serviceability or durability.

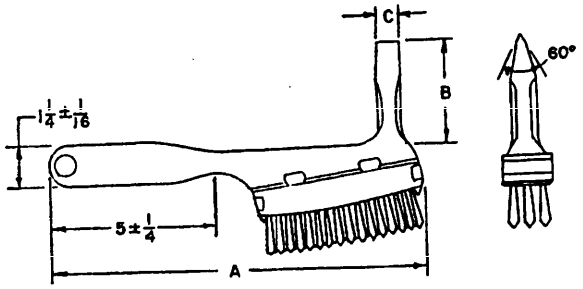


FIGURE 6.—Type IV, straight peen and brush (with style E, forged steel handle).

4. SAMPLING, INSPECTION AND TEST PROCEDURE

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All hammers of the same type, class, and size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of hammers shall be taken by the inspector in accordance with table IV from each inspection lot of material offered for inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements of Military Standard MIL–STD–105.

4.1.3 *Sampling for lot acceptance tests except plastic handle test.*—A random sample of hammers shall be taken by the inspector from each inspection lot in accordance with table V and shall be subjected to the tests specified in 4.3.1.

4.1.4 *Sampling for plastic handle tests.*—A random sample of hammers shall be taken in accordance with table VI and shall be subjected to the plastic handle tests (see 4.4.3).

4.2 **Visual and dimensional inspection.**—Each of the sample hammers taken in accordance with table IV shall be visually and dimensionally inspected by the inspector to verify compliance with this specification. Any hammer in the sample containing one

TABLE IV.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of hammers in inspection lot	Number of hammers in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1,300	75	3	4
1,301 and over	110	4	5

Note.—The inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL–STD–105.

TABLE V.—*Sampling for lot acceptance tests*

Number of hammers in inspection lot	Number of hammers in sample	Number of items nonconforming on any lot acceptance test	
		Acceptance number	Rejection number
40 and under	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	2	3
801 to 1,300	35	3	4
1,301 and over	50	4	5

TABLE VI.—*Sampling for plastic handle tests*

Number of hammers in inspection lot	Number of hammers in sample
110 and under	2
111 to 500	3
501 and over	5

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or more visual or dimensional defects shall be rejected, and if the number of defective hammers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Lot acceptance tests.**

4.3.1 *Physical and hardness test.*—Each of the sample hammers taken in accordance with 4.1.3 shall be subjected to the tests specified in 4.4.1 and 4.4.2 to verify compliance with this specification. Any sample hammer which fails to pass these tests shall be rejected, and if the number of nonconforming hammers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Plastic handle tests.* — Each of the sample hammers taken in accordance with 4.1.4 shall be subjected to the tests specified in 4.4.3 to verify conformance with this specification. If any sample hammer fails the test the lot shall be rejected.

4.4 **Test procedures.**

4.4.1 *Physical tests.*—The hammer shall be held firmly in the hand and a blow shall be struck as near to the vertical as possible on an annealed carbon-steel plate (0.80 to 0.90 percent carbon) having a minimum width of 3 inches, a minimum thickness of ¾ inch, and of sufficient length to clamp securely in a rigid manner upon a solid foundation. The plate shall have a hardness value of not less than 35 nor more than 39 on the Rockwell C scale. The blow shall be struck with sufficient force to produce a minimum indentation of $\frac{1}{32}$ inch. After making five such indentations with each chipping end of the hammer, the cutting edge of the chisel end and the point of the punch end shall be inspected to determine conformance with 3.3.1.1.

4.4.1.1 *Weldment inspection.*—Weldments shall be inspected following the physical test (see 4.4.1). Weldments shall be free from cracks, fractures, or any other defect which may cause turning or loosening of the head or handle.

4.4.2 *Hardness test.*—Hardness test shall be in accordance with Federal Test Method Standard No. 151 and the value thus obtained shall comply with 3.3.1.1.1. Impressions shall be taken within ½ inch of each working end of the head.

4.4.3 *Plastic handle tests.*

4.4.3.1 *Flammability test.* — The round mild-steel bar of the handle for test shall be clamped in a support with the plastic handgrip extended outward in a horizontal position. A piece of 10-mesh Bunsen burner gauze about 5 inches square shall be clamped under the plastic handgrip in a horizontal position, ¼ inch below the bottom of the handgrip and with about ½ inch of the handgrip extending beyond the edge of the gauze. A Bunsen burner with an efficient flame of ½ to ¾ inch in height shall be adjusted so that the flame tip will just contact the handgrip. The burner shall be placed under the handgrip for 30 seconds and then removed. The handgrip shall be allowed to burn. The plastic handgrip shall not burn at a rate exceeding 2 inches per minute. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handgrip after the removal of the burner.

4.4.3.2 *Heat distortion test.*—The plastic handgrip of the handle under test shall be completely immersed in boiling water for 2 minutes. The handgrip, after being removed from the boiling water, shall show no significant distortion or blistering.

4.5 **Rejected lots.**—Rejected lots may be offered again for inspection provided the contractor has repaired or removed all nonconforming hammers. The inspector shall again take and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1).

5.1 **Packaging.**

5.1.1 *Level A.*—Level A cleaning, preser-

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vation, and packaging shall be in accordance with level A in Military Specification MIL–P–15424.

5.1.2 *Level C.*—Level C preservatives shall be applied in accordance with the contractor's commercial practice. Level C unit protection and intermediate containers, if used, shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Level A packing shall be in accordance with level A in Military Specification MIL–P–15424.

5.2.2 *Level B.*—Level B packing shall be in accordance with level B in Military Specification MIL–P–15424.

5.2.3 *Level C.*—Level C packing of hammers, preserved and packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.2) shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking for shipment.

5.3.1 *Military.*—Marking for shipment to Military activities shall be in accordance with Military Specification MIL–P–15424.

5.3.2 *Nonmilitary.*

5.3.2.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name of the contractor, or trade-mark of such known character that the source of supply may be readily determined.

5.3.2.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (see 1.2).
(*c*) A particular handle style if required (see 3.3.2).
(*d*) Level of preservation, packaging and packing (see section 5).

6.3 Transportation description. — The proper transportation description is as follows:

Hammers, hand, not otherwise indexed by name, with or without handles.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Copies of this specification may be purchased for 10 cents each.

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